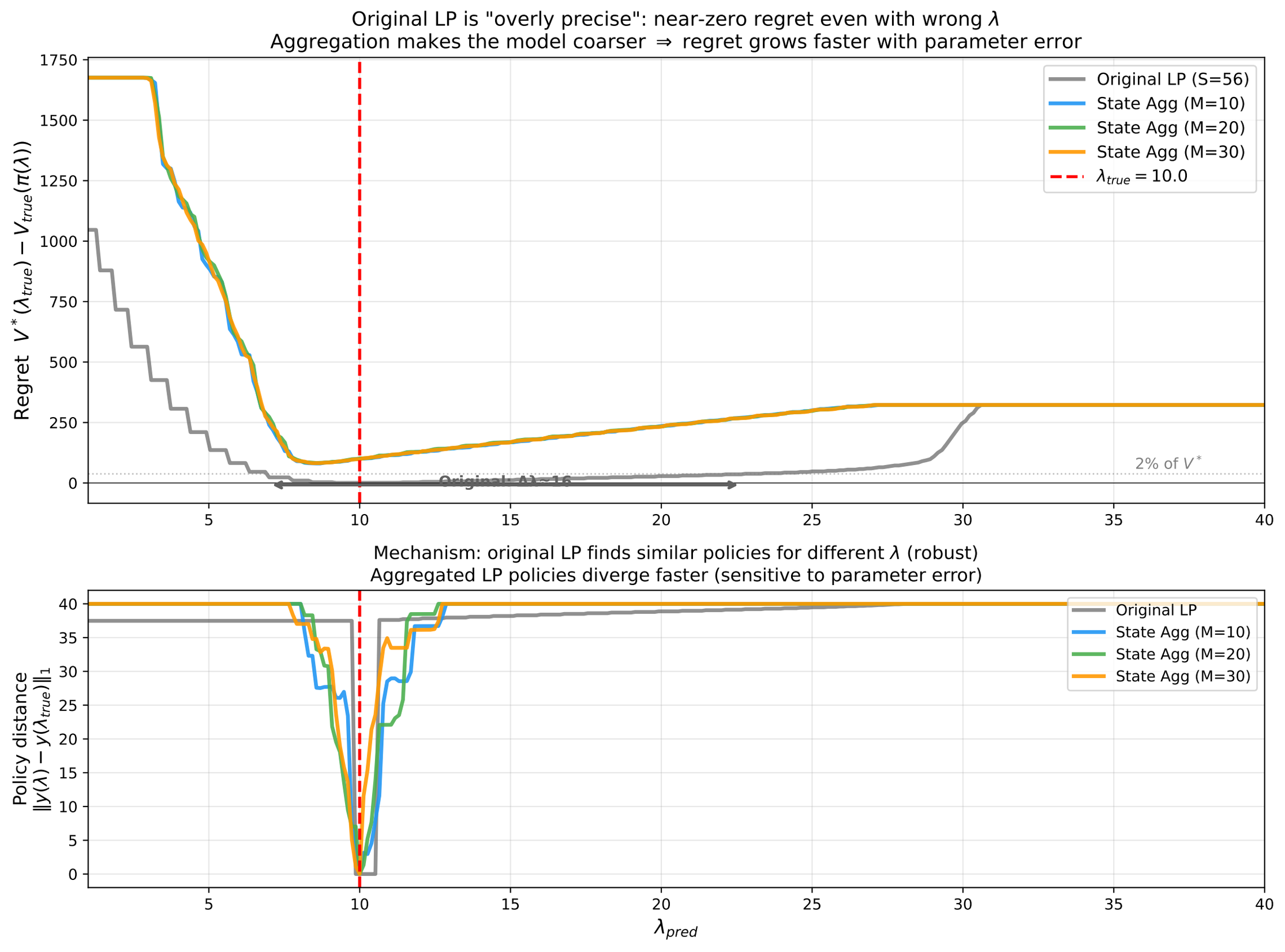
Original LP is "overly precise": near-zero regret even with wrong λ
Aggregation makes the model coarser ⇒ regret grows faster with parameter error
Regret $V^*(\lambda_{true}) - V_{true}(\pi(\lambda))$
Original LP (S=56)
State Agg (M=10)
State Agg (M=20)
State Agg (M=30)
$\lambda_{true} = 10.0$
2% of V^*
Original: Δλ ≈ 16
Mechanism: original LP finds similar policies for different λ (robust)
Aggregated LP policies diverge faster (sensitive to parameter error)
Policy distance
$\|y(\lambda) - y(\lambda_{true})\|_1$
Original LP
State Agg (M=10)
State Agg (M=20)
State Agg (M=30)
λ_{pred}